Paleomagnetism of the Monteregian Hills: Further New Results

A. LAROCHELLE

Geological Survey of Canada, Ottawa, Ontario, Canada

Magnetization measurements were made on a hundred-twelve independently oriented cores drilled at twenty-two sites distributed among six basic intrusive plugs forming part of the Monteregian Hills group in southeast Quebec. On the basis of adopted magnetization homogeneity criteria, less than 80% of the sampled cores were accepted as reliable carriers of paleomagnetic information, after af cleaning in a peak demagnetizing field of 250 oe. An analysis of the data obtained from these cores representative of sixteen sites yielded a mean direction of magnetization whose declination and inclination are, respectively, 157.4° and −59.9°. The circle of confidence about the resultant of the sixteen 'weighted' site means has a radius $\alpha'_{.05}$ equal to 3.3°. No reliable evidence of mixed polarity in the paleomagnetic directions within any one of the six intrusives studied was found, although additional evidence was obtained to postulate at least one earth's field reversal during their formation. Combining the above mentioned data with seventy-five previously reported core mean directions representative of sixteen sites on four other members of the Monteregian Hills group yielded a paleomagnetic pole position at 170.5°W, 71.3°N. The low angular standard deviation of the thirty-two 'weighted' site means summarizing these data (7.8°) suggests that the non-dipole component of the earth's magnetic field was less important in Cretaceous time than it was over the last 10 m.y.

INTRODUCTION

The earth's magnetic field polarity reversal about the time the Monteregian Hills were formed (Lower Cretaceous) was recently confirmed by new paleomagnetic directions [*Larochelle*, 1968*a*] obtained from four of these eight basic intrusive plugs located in southeast Quebec. Some of these data also appeared to indicate, although very inconclusively, the occurrence of both normal and reverse paleomagnetic directions within one of the sampled bodies, Mount Shefford. The presence of both directions would mean that a complete reversal of the earth's field could have taken place within the few hundred years required for relatively small sectors of Mount Shefford to cool through the Curie temperature of their ferromagnetic constituents. One could then conclude that the process of polarity reversals was considerably faster in Cretaceous time than it is believed to have been a few million years ago, according to recent studies [*Cox and Dalrymple*, 1967]. Also, the paleomagnetic pole position recently obtained for the Monteregian Hills that were studied conforms remarkably well with poles obtained from other Cretaceous but far distant North American formations. It seemed pertinent to verify whether this result could be extended to the other four Monteregian Hills and to their apparent extensions, the Mount Megantic and Oka intrusives, which are located beyond the perimeter of the Monteregian Hills proper (Figure 1). Paleomagnetic directions were obtained from the six intrusives last referred to; this paper gives a description and an analysis of these data.

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SAMPLING AND LABORATORY PROCEDURES

One hundred and twelve cores were drilled from twenty-two sites distributed among the six intrusive bodies referred to as St. Hilaire, Rougemont, Yamaska, Royal, Megantic, and Oka (Figure 1). Within each of these bodies, the sampling sites were spaced between 1 and 3 km apart with the exception of sites H-3-A and Y-5-A, which are less than ½ km away from sites H-3 and Y-4, respectively (Table 1).

At each site, a minimum of five cores were drilled between 3 and 5 meters apart and oriented in situ with a solar compass aligned on the horizontal plate of a sampling tube that incorporates a clinometer. The orientation error is estimated to be about 2°.

From each core, two cylinders (3.2 cm in

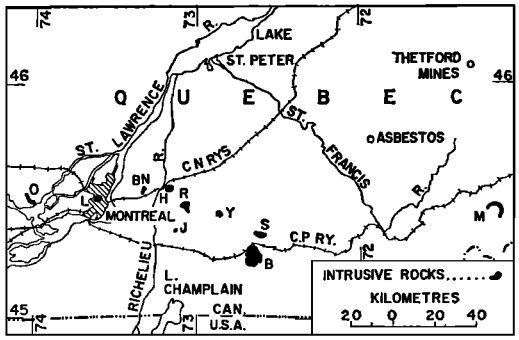


Fig. 1. Areal distribution of Monteregian Hills intrusives and extensions. O, Oka; L, Royal; BN, St. Bruno; H, St. Hilaire; R, Rougemont; Y, Yamaska; S, Shefford; B, Brome; M. Megantic; J, Johnson.

diameter and 2.8 cm long) were cut and the natural remanent magnetization (NRM) of each was measured with a biastatic magnetometer whose limit of resolution is 0.5° or less [*Larochelle and Christie*, 1967].

Each cylinder was submitted to an alternating-field (af) cleaning treatment with the apparatus described by *Larochelle and Black* [1965]. The peak demagnetizing field intensity was set at 250 oe on the basis of previous stability tests on Monteregian Hills rocks [*Larochelle*, 1968*a*]. The remanent magnetization of each cylinder was then remeasured and, for each core, the mean direction of magnetization was taken as that of the bisectrix of the two af cleaned magnetization vectors obtained for it. The angle θ between these two vectors was used as a measure of the within-core magnetization homogeneity. Cores for which θ exceeded 20° were not considered as reliable carriers of paleomagnetic information in the present context. This sets to 10° the maximum standard error $\delta_{\bar{c}}$ for the magnetization direction of a reliable core, since the standard error is defined in the present case as

$$\delta_{\bar{c}} = [2 - 2 \cos(\theta/2)]^{1/2} \quad \text{radians}$$

Results

A histogram of θ (Figure 2) indicates that less than 80% of the cores were apt to yield reliable paleomagnetic information, according to the above specifications. In the light of past experience, this appears to be a normal recovery for basic intrusive rocks and fully justifies estimating the magnetization direction of a sample (core) from the directions obtained for at least two specimens (cylinders) when dealing with this rock type. The elimination of data obtained for inhomogeneously magnetized cores will obviously contribute to reduce to more realistic proportions the angular standard deviation of a set of paleomagnetic directions; this is critical when attempts are made to evaluate, on this basis, the scope of the magnetic field secular variation in the geological past.

The mean directions of magnetization of cores considered reliable were grouped into sites and the site means were determined vectorially. The standard errors of the site means, $\delta_{\bar{E}}$, were computed according to the equation

$$\delta_{\bar{E}} = [2(n - R)/n(n - 1)]^{1/2} \quad \text{radians}$$

TABLE 1. Site Mean Directions

Intrusive	Site	N	n	D, deg	I, deg	$\delta_{\bar{E}}$, deg
St. Hilaire	H-1	5	2	125.2	−49.5	34.2
	H-2	5	5	155.6	−59.3	0.9
	H-3	5	2	144.1	−52.2	12.7
	H-3A	5	0			
Yamaska	Y-4	5	3	320.0	47.0	6.4
	Y-4A	5	5	147.3	−69.8	2.9
	Y-5	5	5	156.6	−58.0	2.9
	Y-5A	5	5	158.3	−58.6	1.9
	Y-6	5	5	162.6	−61.2	2.6
Rougemont	R-7	5	4	326.6	21.7	31.4
	R-8	6	3	85.5	−63.6	41.5
	R-9	5	5	169.2	−53.5	4.4
Megantic	*M-10*	5	5	154.3	−46.0	3.4
	M-11	5	5	148.7	−62.4	2.6
	M-12	5	0			
Royal	L-13	6	5	115.6	−67.2	12.7
	L-14	5	5	147.6	−58.0	2.9
	L-15	5	5	184.5	−65.3	2.3
	L-16	5	5	135.3	−71.6	12.6
Oka	*O-17*	5	4	331.7	54.3	5.5
	O-18	5	4	345.9	59.4	5.3
	O-19	5	4	340.8	60.0	3.8

Notes.
N, total number of cores drilled at a site.
n, number of cores for which $\theta \leq 20°$.
D, I, declination and inclination of site mean based on n cores.
$\delta_{\bar{E}}$, standard error of site mean.

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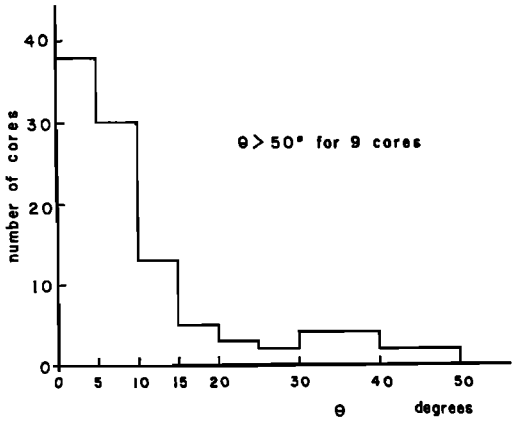


Fig. 2. Histogram of θ (see text).

where n is the number of reliable directions and R is the modulus of their resultant at a given site.

The results of these computations (Table 1) show that the value of $\delta_{\bar{E}}$ was smaller than 10° for the majority of the sites. On the other hand, one should normally reject as unreliable the site means for which $\delta_{\bar{E}}$ exceeds 10° if the same norms of homogeneity are to be accepted at the site level of sampling as at the core level. This rule was applied, obviously for sites H-3-A and M-12, where none of the five sampled cores were found reliable, and for sites H-1 and H-3, where only two of the five cores appeared to be reliable. A closer examination of the apparently reliable directions obtained for sites R-7, R-8, L-13, and L-16 revealed, however, that the abnormally high values of $\delta_{\bar{E}}$ obtained for these sites hinged in each case on one erratic direction. An example of this is in the data given in Table 3 for site R-7. The magnetization direction of core R-7-4 was found to diverge by an angle ψ of 145° from the mean of the other three reliable vectors (D', I') whose standard error, $\delta_{\bar{E}}'$, is 7.1°. As the value of ψ is substantially greater than $[3(n-1)^{1/2}]\,\delta_{\bar{E}}'$, it can be shown that the probability of the erratic vector belonging to the same population as the other three reliable ones is considerably less than 0.001. Accordingly, the adjusted statistics D', I', and $\delta_{\bar{E}}'$ were considered more representative than the values given in Table 1 for site R-7, and they were utilized preferentially throughout the rest of this analysis. The same preference was given to the corresponding adjusted statistics listed in Table 2 for sites R-8, L-13, and L-16.

It is noted in Table 1 that the three site means obtained for the Oka intrusive are almost antiparallel to the majority of the other site means. This was interpreted as additional evidence for postulating at least one polarity reversal of the earth's magnetic field during the formation of the Monteregian Hills and their extensions. On the other hand, the downward-pointing mean direction of magnetization obtained for site Y-4 might suggest the possibility that the earth's field polarity was reversed during the relatively short period separating the cooling of sites Y-4 and Y-5-A, respectively. As mentioned earlier, these two sites are less than ½ km apart. A close examination of the data obtained from site Y-4 (Table 3) suggests, however, that the downward NRM polarity of the cores could be attributed to a stable component of chemical magnetization recently acquired by the rocks as a result of weathering. This would explain the consistency of the NRM data along the general direction of the present earth's field at site Y-4, in contrast with the high dispersion of NRM directions normally observed at most sites [*Larochelle*, 1968*a*]. This high dispersion is most likely due to soft components of viscous magnetization acquired by samples after their removal from the outcrop and prior to their measurement in the laboratory. At site Y-4 the postulated chemical component would effectively mask the viscous as well as the original thermoremanent components and it would continue to do so more or less efficiently even after the magnetic cleaning treatment. Thus, an explanation would be provided for the systematic decrease in the positive

TABLE 2. Adjusted Site Mean Directions

Site	N	n'	D', deg	I', deg	$\delta_{\bar{E}}'$, deg	ψ, deg
R-7	5	3	326.6	36.6	7.1	145
R-8	6	2	155.6	−66.0	4.0	136
L-13	6	4	138.2	−61.3	4.7	63
L-16	5	4	158.3	−63.9	2.6	64

Notes.

D', I', $\delta_{\bar{E}}'$, 'adjusted' statistics of site mean (see text).

ψ, deviation of erratic vector from the mean of the other n' vectors.

Other symbols as for Table 1.

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TABLE 3. Details of Data from Sites Y-4 and R-7

Core	NRM			After 250-oe Cleaning		
	D, deg	I, deg	θ, deg	D, deg	I, deg	θ, deg
Y-4-1	344.3	65.7	3.3	326.3	39.9	18.6
Y-4-2	335.9	65.7	3.1	327.9	49.5	4.7
Y-4-3	14.5	68.4	1.3	41.2	−82.4	51.3
Y-4-4	36.5	72.1	3.3	1.9	55.0	26.5
Y-4-5	7.1	67.1	1.9	328.8	41.6	8.2
R-7-1	141.3	26.7	106.8	183.4	−80.0	32.8
R-7-2	24.0	64.9	3.6	340.3	30.8	6.7
R-7-3	353.8	52.6	0.9	318.4	33.4	3.8
R-7-4	21.1	−16.4	11.6	146.4	−71.1	0.7
R-7-5	351.0	38.8	8.0	319.9	44.2	13.1

Notes.

θ, angle between magnetization vector of two cylinders representative of a core.

Other symbols as for Table 1.

magnetic inclinations of all cores from this site after the magnetic cleaning treatment. This is particularly noticeable for one of the cores, which even became reversely polarized, although inhomogeneously, after the treatment. The normal polarity obtained for the mean of site R-7 can be explained in almost the same terms, except that in this case, the polarity of two cores was reversed after the cleaning treatment and one of them even became very consistently magnetized along the general paleomagnetic direction of the Monteregian Hills. It seems therefore difficult to regard the data obtained from sites Y-4 and R-7 as conclusive evidence indicating that the earth's field had its polarity reversed during the partial cooling of the Yamaska and the Rougemont intrusives. The data obtained for sites Y-4 and R-7 were not considered further in the rest of this study because of their unreliability.

Statistical Analysis

The data so far considered reliable are summarized as site means listed opposite site numbers printed in *italics* in Tables 1 and 2.

The within-site (δ_w^2) and the between-site (δ_b^2) angular variances [*Larochelle*, 1968*b*] were first computed for each of the intrusives except St. Hilaire, where only one out of four sites sampled yielded reliable information. Table 4 shows that the δ_b^2/δ_w^2 ratio is appreciably larger than the corresponding theoretical statistic $F_{.05}$ for the Yamaska, Royal, and Megantic intrusives but that it is smaller in the case of Oka and Rougemont. The results of the test are therefore inconclusive as to whether or not the site means should be considered as significantly distinct.

After the polarity of the Oka intrusive data was inverted, the seventy-two core mean direc-

TABLE 4. Results of Variance-Ratio Tests

Intrusive	n	B	D, deg	I, deg	δ, deg	δ_b^2, rad^2	δ_w^2, rad^2	δ_b^2/δ_w^2	$F_{.05}$	δ_m, deg	δ_n, deg
Yamaska	20	4	156.9	−62.0	7.6	.05620	.01040	5.40	2.40	6.1	6.1
Rougemont	7	2	166.2	−57.2	11.0	.08824	.02629	3.36	4.10	9.1	10.1
Megantic	10	2	152.0	−54.3	10.8	.21194	.01355	15.65	3.63	11.8	11.8
Royal	18	4	156.7	−63.1	10.6	.13009	.01349	9.64	2.44	9.7	9.5
Oka	12	3	339.1	58.0	9.8	.03044	.02926	1.04	2.93	5.0	5.0
Six intrusives	72	16	157.4	−59.9	9.8	.08030	.01541	5.21	1.56	7.6	7.5
Six intrusives	(16)	(6)	157.2	−60.2	7.5	.01068	.02049	0.52	2.35	3.6	3.7
Ten intrusives	147	32	155.5	−59.2	10.2	.08415	.01789	4.70	1.38	7.8	7.8

Notes.

n, number of core mean directions (site means).

B, number of sites (intrusives).

D, I, mean of n directions.

δ, angular standard deviation of n directions.

δ_w^2, δ_b^2, within-site (intrusive) and between-site (intrusive). angular variances [*Larochelle*, 1968*a*, *b*].

$\delta_m = \delta_b(B/n)^{1/2}$ radians: 'weighted' site means angular standard deviation.

δ_n, angular standard deviation of 'normalized' site means.

$F_{.05}$, theoretical value of $F_{2(B-1),2(n-B),.05}$.

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tions were grouped by sites, independently of intrusives, and another variance-ratio test showed clearly that the site means are significantly different throughout the six intrusives studied. The standard deviation of the sixteen 'weighted' site means is defined in the present case by

$$\delta_m = \delta_b(16/72)^{1/2} = 7.7°$$

and the standard error of their resultant (157.4°, −59.9°) is given by

$$\delta_{\bar{m}} = \delta_m/(16)^{1/2} = 1.9°$$

As the more familiar statistic $\alpha'_{.95}$ is actually equal to $\sqrt{3}\,\delta_{\bar{m}}$, its value is 3.3° in the present case.

The sixteen 'normalized' site means were then grouped by intrusives, and their within- and between-intrusive angular variances were computed. The variance-ratio test again indicated clearly (Table 4) that the intrusive means are not significantly different, as had been found previously for the four other intrusives of the group [*Larochelle*, 1968*a*].

Incorporation of Earlier Data

The core mean directions obtained earlier for the St. Bruno, Shefford, Brome, and Johnson intrusives [*Larochelle*, 1968a] were previously considered reliable when their characteristic angle θ was less than 15°. If this tolerance is raised to 20°, in conformity with the present context, four previously rejected core mean directions are reintegrated as valid and the means of sites 4, 9, and 10 are shifted slightly from their reported positions. Nevertheless, the mean of site 9 is still found unacceptable because of its prohibitively high standard error (19.3°) and the mean of site 4 remains unreliable for reasons identical to those given above in rejecting the data from sites Y-4 and R-7. The recalculated mean of the seventy-five directions representative of the four intrusives (153.7°, −58.5°) is in effect almost the same as that obtained earlier (154.0°, −58.5°) with slightly more rigid criteria of acceptance.

The sixteen site means summarizing the data just referred to and the sixteen site means reported in this study were grouped as two samples, the means of which were compared by the variance-ratio test. As the δ_b^2/δ_w^2 ratio (1.284) is appreciably smaller than the statistic $F_{.260,\ .05}$ 3.15), the two samples means were considered to be not significantly different. Furthermore, noting that the two samples have approximately the same standard deviation (8.1° and 7.7°, respectively), one is justified to blend the two sets of data into one.

The 147 core mean directions considered reliable in both studies were then grouped by sites, independently of intrusives, and the between- and within-site angular variances were computed. The thirty-two site means were again found significantly different by the variance-ratio test (Table 4) and their weighted standard deviation ($\delta_m = 7.8°$) was accepted as a maximum estimate of the earth's field angular secular variation at the time the Monteregian Hills were formed. The resultant of the thirty-two weighted site means (155.5°, −59.2°) has a standard error ($\delta_{\bar{m}} = \delta_m/\sqrt{32}$) of 1.4°, and it thus centers a circle of confidence of radius $\alpha'_{.95}$ equal to 2.4°. On the other hand, the resultant of the thirty-two normalized site means (155.2°, −59.0°) has the same standard error ($\delta_{\bar{n}} = \delta_n/\sqrt{32}$) and radius of confidence as the resultant of the weighted site means.

By using 72.80°W, 45.30°N, as the mean coordinates of the Monteregian Hills, their paleomagnetic pole position was computed on the basis of the resultant of the thirty-two weighted site means. The choice of this basis lies in the fact that the site means have somewhat unequal standard errors and that assigning equal weight to all site means may be misleading in certain cases. The pole position (170.5°W, 71.3°N) is about 1.5° from the position reported previously on the basis of data obtained from only four of the ten intrusives [*Larochelle*, 1968*a*].

Discussion

A significant result of the above analysis is the low angular standard deviation of the site means which characterizes the Monteregian Hills data. This suggests that the non-dipole component of the earth's magnetic field must have been less important in Cretaceous time than during the last few million years, judging from the data obtained by *Cox and Doell* [1964] from western Pacific lavas of Recent age. Furthermore, considering that some of the sites may have undergone minor tilting since

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their emplacement and that the paleomagnetic field was not necessarily free of local anomalies over the 150-km span that the thirty-two sites cover, the angular standard deviation of the present paleomagnetic data must be considered as a maximum estimate of the paleomagnetic field secular variation.

It might be argued that the formation of the Monteregian Hills may have been an event of short duration, so that the above data cannot be used as a reliable gage of the secular variation in Cretaceous time. It is pointed out, however, that the group of intrusives has witnessed at least one earth's field reversal during its cooling, and, if it is correct to estimate with *Cox and Dalrymple* [1967] that this phenomenon spreads over about 5000 years, the Monteregian Hills may have taken that long to cool through the Curie point of magnetite. In the perspective of the numerous Rb/Sr and K/Ar age determinations reported by *Fairbairn et al.* [1963] on the Monteregian Hills, it appears, however, that the group of intrusives may in fact have taken several million years to form.

That the earth's field was apparently entirely dipolar in Cretaceous time is also suggested by the apparently reliable pole positions derived from other widely separated North American formations of this period. The high clustering of these poles about their center of gravity [*Larochelle*, 1968*a*] suggests that the 'wobble' of the dipole at that time was much less pronounced than it has been over the last 10 m.y., although further evidence is obviously required to confirm this view.

Summary and Conclusions

The practice of measuring magnetization directions of two specimens (cylinders) from each sample (core) to evaluate the paleomagnetic direction of the sample no longer appears to be a useless precaution when dealing with basic intrusive rocks. The angle θ between the two directions thus obtained provides a reliable index of the within-core homogeneity of magnetization. Ultimately, when inhomogeneously magnetized samples are ignored, a more realistic estimate of the angular standard deviation of the paleomagnetic field may be obtained from a group of paleomagnetic data, although this estimate must still be considered as a maximum.

The data reported as reliable in the present study fully confirmed the Cretaceous pole position previously derived from only four of the Monteregian Hills. Also, as previously determined, no valid evidence was obtained to indicate paleomagnetic directions of mixed polarities within any one intrusive, but the reversal of the Monteregian Hills paleomagnetic field was confirmed by additional data.

The result of the statistical analysis indicating that the intrusive means are not significantly different strongly supports the view that the Oka and Mount Megantic intrusives are true extensions of the Monteregian Hills proper.

Finally, the low angular standard deviation of the analyzed data suggests that in Cretaceous time the non-dipole component of the earth's magnetic field was less important relative to the dipole field than it is today.

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